Question	Answer	Marks	Partial Marks
1(a)(i)	4.55 or 4.545…	**1**	
1(a)(ii)	50 : 263 : 400 cao	**2**	**M1**for a correct simplification from 250 000 : 1 315 000 : 2 000 000
1(a)(iii)	83 cao	**3**	**M2** for $\frac{43\frac{1}{3}}{100}\times(100-60-10)$ oe or **M1** for 100 – 60 – 10 seen
1(a)(iv)	10 200 000 cao	**3**	**B2** for 10 185 185 to 10 185 200 or **M1** for 5 500 000 ÷ 27 [× 50]
1(a)(v)	3.19×10^7 or $3.190\ldots \times 10^7$	**3**	**B2** for 31903920 or **M1** for 60.7 × 60 × 24 × 365 If B0 scored **SC1** for correctly converting *their* number seen to standard form to 3sf or better
1(b)	2095 nfww	**3**	**M2** for 6445 – C where $4300 \leqslant C < 4400$ oe or A – 4350 where $6440 < A \leqslant 6450$ oe or **M1** for 6440 +5 or 6440 –5 or 4400 + 50 or 4400 – 50 seen oe
2(a)(i)	Triangle at (2, 1) (1, 3) (5, 3)	**1**	
2(a)(ii)	Triangle at (–4, –5) (–3, –3) (0, –5)	**2**	**B1** for translation by $\begin{pmatrix}-5\\k\end{pmatrix}$ or $\begin{pmatrix}k\\-2\end{pmatrix}$
2(a)(iii)	Triangle at (–2.5, 2) (–4, 3) (–2, 3)	**2**	**B1** for enlargement by sf $-\frac{1}{2}$ with any centre
2(b)	14.4	**3**	**M2** for [10 ×] $3^2 \times \left(\frac{2}{5}\right)^2$ oe or **M1** for 3^2 or $\left(\frac{2}{5}\right)^2$ soi
3(a)(i)	80	**2**	**M1** for $\frac{1}{4}\times 5\times 8^2$
3(a)(ii)	5	**2**	**M1** for [y^2 =] $\frac{15\times 4}{2.4}$ oe

Question	Answer	Marks	Partial Marks
3(b)	$\frac{5x+23}{(x-1)(2x+5)}$ or $\frac{5x+23}{2x^2+3x-5}$ final answer	**3**	**B1** for $4(2x + 5) - 3(x - 1)$ oe isw **B1** for common denominator = $(x - 1)(2x + 5)$ oe isw
3(c)	$2x^3 - 13x^2 + 8x + 48$ final answer	**3**	**B2** for correct expansion of 3 brackets but unsimplified or for simplified four-term expression of correct form with 3 terms correct or **B1** for correct expansion of two brackets with at least 3 terms out of 4 correct
3(d)	$\frac{8x^{12}}{y^6}$ or $8x^{12}y^{-6}$ final answer	**3**	**B2** for two elements correct in final answer or for correct answer seen then spoiled or for correct expression where all parts of the power have been dealt with or for $(\ )^{-1}$ or $\left(\frac{2x^4}{y^2}\right)^3$ or **B1** for 8 or y^6 or y^{-6} or x^{12} correct in final answer or for $\left(\frac{16x^{16}}{y^8}\right)^{\frac{3}{4}}$ or $\left(\frac{y^2}{2x^4}\right)^{-3}$
4(a)(i)	9.3	**1**	
4(a)(ii)	3.4	**1**	
4(a)(iii)	63	**5**	**M4** for $\frac{195}{6}\times\frac{3600}{1000}-\frac{195}{13}\times\frac{3600}{1000}$ oe or **M3** for $\frac{195}{6}\times\frac{3600}{1000}$ oe or $\frac{195}{13}\times\frac{3600}{1000}$ oe or for $(\frac{195}{6}-\frac{195}{13})[\times k]$ oe OR **M1** for $\frac{195}{6}$ or $\frac{195}{13}$ or *their* speed $\times\frac{3600}{1000}$ seen **M1** for selecting 6 and 13